Executive Summary – Néovolt Grid+ Program:

Néovolt is a regional energy distributor operating electricity and gas services for approximately 600,000 households and business customers across both urban and rural areas. As an operator of critical infrastructure, the company faces growing operational, financial, regulatory and cybersecurity challenges. The deployment of smart meters has created a large volume of consumption data, but this data is currently underused, fragmented and affected by quality issues. At the same time, peak demand is insufficiently anticipated, anomalies and fraud are detected too late, business teams rely on scattered Excel files, and the overall security and governance of the data platform remain insufficiently structured. The Néovolt Grid+ program aims to modernize the way Néovolt collects, processes, analyzes, secures and governs its network and consumption data. The objective is to provide a credible prototype and a decision-oriented project dossier that can support the steering committee in deciding the next phase of investment. Our proposed approach focuses on building a coherent data-driven foundation around three priorities. First, consumption and network data must be centralized, cleaned and documented in order to create a trusted analytical base. Second, the project should provide decision makers with clear and reliable dashboards adapted to their needs: network operations, finance and customer relationship management. Third, analytical and machine learning use cases should be developed to support demand forecasting and anomaly or fraud detection, while remaining explainable and subject to human review. The target architecture will remain realistic and compatible with Néovolt's existing information system. The prototype will rely on simple and reproducible tools such as Python, pandas and Power BI. The architecture will not directly interfere with the SCADA environment, which must remain isolated due to its critical role in network supervision. Data hosting and processing choices must respect European data sovereignty requirements and avoid excessive dependence on a single provider. Security, ethics and governance are embedded from the start. Consumption data is personal data and must be processed according to GDPR principles: lawful basis, data minimization, limited retention, access control and transparency. Fraud detection must not lead to fully automated adverse decisions against customers. Any suspicious case should be explainable and reviewed by a human operator before action is taken. The project also integrates data governance principles, including data ownership, quality rules, traceability, access rights and lifecycle management. The expected business value is significant. Better demand forecasting can reduce emergency energy purchasing costs and improve grid stability. Faster anomaly and fraud detection can reduce non-technical losses. Improved dashboards can help operational, financial and customer service teams make faster, better-informed decisions. However, these benefits must be confirmed through further analysis of the provided datasets and through a controlled industrialization phase. The recommended next step is to validate the prototype on the provided datasets, assess the quality and reliability of the results, prioritize the most valuable use cases, and define a phased roadmap for industrial deployment. This roadmap should include governance, cybersecurity, change management, monitoring of models over time, and a realistic budget aligned with Néovolt's constraints.